Macroeconomics I — MPE Insper

Practice list: The New Keynesian AS–AD Model — graduate level (Lista 7)

Solutions and grading rubric

Instructions. Multiple choice: mark the single correct option (no penalty for wrong answers). *True/False*: every 2 wrong items cancel 1 right one (score = $\max\{0, \text{right} - \text{wrong}/2\}\times$ item value); blank items do not count. Discursive questions: show every step. Numbers are illustrative and chosen to give clean answers.

Notation (Benigno 2015). Lower-case letters are log deviations from the steady state; bars denote the long run. Households: $u(C) - v(L) + \beta\{u(\bar{C}) - v(\bar{L})\}$, $u = C^{1-\tilde{\sigma}^{-1}}/(1-\tilde{\sigma}^{-1})$, $v = L^{1+\eta}/(1+\eta)$, $\rho \equiv -\ln\beta$. Resources: $y = s_c c + g$, $\sigma \equiv \tilde{\sigma} s_c$. Firms: demand $Y(j) = (P(j)/P)^{-\theta}(C+G)$, technology $Y(j) = AL(j)$; a fraction α of prices is preset at P^e. Calibration used throughout unless stated: $\alpha = 0.75$, $\sigma = 0.5$, $\eta = 0.2$, $\theta = 8$, $\rho = 0.01$ (a quarter), $p^e = \bar{p} = 0$, $g = \bar{g} = 0$, no taxes.

Block 1 — Foundations: AD, AS, natural and efficient output

(Benigno §§1–5 • 25 points)

Question 1. (10 points) Objective items.

a) (3 points) The slope κ of the AS curve $p - p^e = \kappa(y - y_n)$ under the calibration is:

A. 6.600

B. 0.733

C. 0.233

D. 2.200

Answer: B

Step — derivation. Flexible firms price at $\tilde{p} = w - a + \mu$; labor supply gives $w - p = \eta l + \sigma^{-1}(y - g)$ with $l = y - a$; so $\tilde{p} - p = (\sigma^{-1} + \eta)(y - y_n)$. With $p = \alpha p^e + (1-\alpha)\tilde{p}$: $\kappa = \frac{(1-\alpha)(\sigma^{-1}+\eta)}{\alpha} = \frac{0.25 \cdot 2.2}{0.75} = 0.733$ (Question 2(b) in full).

Step — distractors. A inverts $(1-\alpha)/\alpha$; C uses $\sigma + \eta$; D omits the rigidity factor.

Ref: Benigno (2015), §4, eqs. (16)–(17), pp. 506–508.

b) (3 points) Productivity rises by $a = 0.02$ and the markup by $\mu = 0.05$, both temporarily. The changes in $(y_n,\ y_e)$ are:

A. $(-0.0227,\ 0)$

B. $(0.0109,\ 0.0109)$

C. $(-0.0118,\ -0.0118)$

D. $(-0.0118,\ 0.0109)$

Answer: D

Step — coefficients. $\sigma^{-1} + \eta = 2.2$. From (15): $\Delta y_n = (1.2 \cdot 0.02 - 0.05)/2.2 = -0.0118$. From (19): $\Delta y_e = 1.2 \cdot 0.02/2.2 = 0.0109$. The markup moves only y_n.

Ref: Benigno (2015), §4.4, eqs. (15) and (19), pp. 508–509.

c) (2 points, T or F) Because θ sets the markup, a higher demand elasticity θ makes the AS curve steeper.

Answer: F

Step — why. θ enters μ, hence y_n (the AS shifts), but not $\kappa = (1-\alpha)(\sigma^{-1}+\eta)/\alpha$: with linear technology and an economy-wide labor market there is no real rigidity. θ reappears in the welfare weight of the loss function (Question 4).

Ref: Benigno (2015), §4, eq. (17); Romer (2012), §6.5–6.7, pp. 268–280.

d) (2 points, T or F) Holding i, $\bar{p}$ and p^e fixed, news that long-run government spending will be higher ($\bar{g} \uparrow$, g unchanged) lowers current output.

Answer: T

Step — channel. In the AD, $\bar{y}_n - \bar{g}$ falls by $\frac{\eta}{\sigma^{-1}+\eta}\Delta\bar{g}$ (long-run consumption falls, as higher future taxes make households poorer); consumption smoothing lowers current demand. The AS does not move, so y falls by $m_{\bar{g}}\Delta\bar{g}$, $m_{\bar{g}} = \eta/D = 0.067$, $D = (\sigma^{-1}+\eta)(1+\kappa\sigma)$.

Ref: Benigno (2015), §8, Table 1, p. 515.

Question 2. (15 points) Discursive. Build the model from its primitives (no taxes, τ's $= 0$).

a) (3 points) From the household problem with budget $PC + \frac{\bar{P}\bar{C}}{1+i} = WL + \frac{\bar{W}\bar{L}}{1+i} + T$, derive the Euler equation and the labor-supply condition. Log-linearize the Euler equation and combine it with $y = s_c c + g$ to obtain the AD curve. Give its slope in the (y, p) plane and its shifters.

Step — FOCs. Lagrangian with multiplier λ on the budget: $u_c(C) = \lambda P$, $\beta u_c(\bar{C}) = \lambda\bar{P}/(1+i)$, $v_l(L) = \lambda W$. Hence

$$\frac{u_c(C)}{\beta u_c(\bar{C})} = (1+i)\frac{P}{\bar{P}} \equiv 1 + r, \qquad \frac{v_l(L)}{u_c(C)} = \frac{W}{P}.$$

Step — log-linear Euler. $u_c = C^{-1/\tilde{\sigma}}$: $-\frac{1}{\tilde{\sigma}}c + \frac{1}{\tilde{\sigma}}\bar{c} - \ln\beta = \ln(1+i) + p - \bar{p}$. With $\ln(1+i) \approx i$ and $\rho = -\ln\beta$: $\bar{c} - c = \tilde{\sigma}[i - (\bar{p} - p) - \rho]$.

Step — AD. $c = (y-g)/s_c$, $\bar{c} = (\bar{y} - \bar{g})/s_c$; multiply by s_c and use $\sigma = \tilde{\sigma}s_c$:

$$y = \bar{y}_n + (g - \bar{g}) - \sigma[i - (\bar{p} - p) - \rho],$$

since long-run output is the flexible-price $\bar{y}_n$. Slope $dp/dy = -1/\sigma$: a higher p lowers expected inflation $\bar{p} - p$, raises the real rate and current saving. Shifters: i, g, $\bar{g}$, $\bar{p}$ and $\bar{y}_n$ (i.e. $\bar{a}$, $\bar{\mu}$, $\bar{g}$).

Ref: Benigno (2015), §3, eqs. (1)–(6) and (21), pp. 505–509.

Grading rubric.

- FOCs: 1 pt.
- Log-linear Euler with ρ and the real rate: 1 pt.
- AD, slope and shifters: 1 pt. Slope justified by money demand: 0 for this line.

b) (4 points) From the firm's problem derive the optimal flexible price, then the relation $\tilde{p} - p = (\sigma^{-1}+\eta)(y - y_n)$ and y_n. With a fraction α of prices preset at p^e (and $p = \alpha p^e + (1-\alpha)\tilde{p}$ to first order), derive the AS curve and κ.

Step — optimal price. $\Pi(j) = (P(j) - \frac{W}{A})(\frac{P(j)}{P})^{-\theta}(C+G)$. FOC in $P(j)$: $(1-\theta)P(j)^{-\theta} + \theta\frac{W}{A}P(j)^{-\theta-1} = 0 \Rightarrow P(j) = \frac{\theta}{\theta-1}\frac{W}{A}$; in logs $\tilde{p} = w - a + \mu$, $\mu \equiv \ln\frac{\theta}{\theta-1}$ (taxes would add to μ, eq. (13)).

Step — real wage. Labor supply $L^{\eta}C^{1/\tilde{\sigma}} = W/P$: $w - p = \eta l + \tilde{\sigma}^{-1}c = \eta(y-a) + \sigma^{-1}(y-g)$, using $l = y - a$ (aggregate labor demand, first order) and $c = (y-g)/s_c$.

Step — gap relation. $\tilde{p} - p = \eta(y-a) + \sigma^{-1}(y-g) - a + \mu = (\sigma^{-1}+\eta)y - (1+\eta)a - \sigma^{-1}g + \mu$. Natural output sets $\tilde{p} = p$:

$$y_n = \frac{(1+\eta)a + \sigma^{-1}g - \mu}{\sigma^{-1}+\eta}, \qquad \tilde{p} - p = (\sigma^{-1}+\eta)(y - y_n).$$

Step — AS. $p - p^e = (1-\alpha)(\tilde{p} - p^e) = (1-\alpha)[(\tilde{p} - p) + (p - p^e)]$, so $\alpha(p - p^e) = (1-\alpha)(\sigma^{-1}+\eta)(y - y_n)$ and $\kappa = \frac{(1-\alpha)(\sigma^{-1}+\eta)}{\alpha} = 0.733$.

Ref: Benigno (2015), §4, eqs. (9)–(17), pp. 506–508; Romer (2012), problem 6.16, p. 311.

Grading rubric.

- Optimal price from the FOC: 1 pt.
- Real wage from labor supply and resources: 1 pt.
- y_n and the gap relation: 1 pt.

- AS and κ: 1 pt. Quoting κ: max. 1 pt in the item.

c) (4 points) Derive efficient output y_e from the planner's problem and prove $y_n - y_e = -\mu/(\sigma^{-1}+\eta)$. Prove that κ does not depend on θ. Give the limits of the AS as $\alpha \to 0$ and $\alpha \to 1$. What subsidy makes the steady state efficient?

Step — planner. $\max u(Y-G)-v(Y/A)$: $u_c(Y-G) = v_l(Y/A)/A$. In logs, $\sigma^{-1}(y-g)+\eta(y-a) = a$, so $y_e = \frac{(1+\eta)a+\sigma^{-1}g}{\sigma^{-1}+\eta}$ and $y_n - y_e = -\frac{\mu}{\sigma^{-1}+\eta}$: the markup is a wedge between the real wage and the marginal product of labor that the planner does not face.

Step — θ and κ. κ is built from α (who adjusts) and $\sigma^{-1}+\eta$ (how marginal cost moves with output); θ enters only the level of the flexible price, i.e. μ. Hence $\partial\kappa/\partial\theta = 0$.

Step — limits. $\alpha \to 0$: $\kappa \to \infty$, a vertical AS at y_n (classical dichotomy, money neutral). $\alpha \to 1$: $\kappa \to 0$, a horizontal AS at p^e (output is demand-determined).

Step — efficient steady state. Need $1+\mu = 1$: with a revenue subsidy, $\frac{\theta}{\theta-1}\frac{1}{1-\tau_y} = 1 \Rightarrow \tau_y = -\frac{1}{\theta-1}$, a 14.3% subsidy at $\theta = 8$. The loss function of Question 4 is approximated around this point.

Ref: Benigno (2015), §4.4, eq. (19), p. 509; eq. (13), p. 507; footnote 21.

Grading rubric.

- Planner FOC and y_e: 1.5 pts.
- $y_n - y_e$ and the wedge reading: 1 pt.
- κ independent of θ, limits: 1 pt.
- Subsidy: 0.5 pt.

d) (4 points) An oil shock raises the markup by $\mu = 0.05$ while productivity falls by $a = -0.02$, both for the short run only. Prove that the equilibrium gap is $y - y_n = \frac{\sigma}{1+\sigma\kappa}[r_n - i + (\bar{p} - p^e)]$ with $r_n = \rho + \sigma^{-1}(\bar{y}_n - y_n)$. Compute y_n, y_e, r_n; the outcome if the bank keeps $i = \rho$; and the i that delivers $p = p^e$. Where is output relative to y_e in each case?

Step — gap formula. Substitute $p = p^e + \kappa(y - y_n)$ into the AD ($g = \bar{g}$): $y = \bar{y}_n - \sigma(i-\rho) + \sigma(\bar{p} - p^e) - \sigma\kappa(y - y_n)$. Subtract y_n from both sides and collect: $(1+\sigma\kappa)(y - y_n) = \bar{y}_n - y_n - \sigma(i-\rho) + \sigma(\bar{p} - p^e) = \sigma[r_n - i + (\bar{p} - p^e)]$.

Step — levels. $y_n = (1.2(-0.02) - 0.05)/2.2 = -0.0336$; $y_e = -0.024/2.2 = -0.0109$; $r_n = 0.01 + 2(0.0336) = 7.73\%$.

Step — $i = \rho$. Gap $= \frac{0.5}{1.367}(0.0773 - 0.01) = 0.0246$; $y = -0.0090$; $p - p^e = 0.733 \cdot 0.0246 = 0.0180$. Output is *above* efficient ($y - y_e = +0.0019$) with prices up 1.8%.

Step — $p = p^e$. Requires a zero gap, $i = r_n = 7.73\%$: then $y = y_n$, 2.3% below y_e.

Economic intuition. A passive bank lets the shock raise prices and lets demand run ahead of the depressed natural level; strict price stability instead pushes output well below what is efficient.

Ref: Benigno (2015), §5 and §7, pp. 509–514; natural rate on p. 513.

Grading rubric.

- Gap formula proved: 1.5 pts.
- y_n, y_e, r_n: 1 pt.
- Both policy outcomes: 1 pt.
- Comparison with y_e: 0.5 pt. Measuring welfare against y_n: 0 for this line.

Block 2 — Policy: optimal rules, the zero bound and deleveraging

(Benigno §§6–12 • 25 points)

Question 3. (10 points) Objective items.

a) (3 points) A temporary cut in consumption taxes of 2 points (like the 2009 IPI cut on cars, modeled as $\tau_c = -0.02$, entering the markup one for one) changes the natural real rate $r_n = \rho + \sigma^{-1}(\bar{y}_n - y_n) + \sigma^{-1}(g - \bar{g}) + (\bar{\tau}_c - \tau_c)$ by:

A. +2.00 points

B. −1.82 points

C. +0.18 points

D. 0

Answer: C

Step — two channels. Supply: $\Delta\mu = -0.02$, so $\Delta y_n = 0.02/2.2 = 0.0091$, lowering r_n by $\sigma^{-1} \cdot 0.0091 = 0.0182$. Demand: $\bar{\tau}_c - \tau_c$ rises by 0.02 (consumption is cheaper today than tomorrow). Net: +0.0018.

Economic intuition. The bank must *raise* i slightly to keep output at its natural level: the demand effect of an intertemporal price change beats the supply effect.

Ref: Benigno (2015), §8 and p. 513.

b) (3 points) With i fixed, a temporary rise in g of 1% of GDP changes output and the gap $y - y_n$ by, respectively:

A. 1.000% and 0.000%

B. 0.909% and 0.000%

C. 0.732% and 0.067%

D. 0.976% and 0.067%

Answer: D

Step — natural output and rate. $\Delta y_n = \frac{\sigma^{-1}}{\sigma^{-1}+\eta} = 0.909$ per unit; $\Delta r_n = \sigma^{-1}(1 - 0.909) = 0.182$.

Step — gap and output. Gap $= \frac{\sigma}{1+\sigma\kappa}\Delta r_n = 0.366 \cdot 0.182 = 0.067$; output $0.909 + 0.067 = 0.976 = m_g$ (Table 1: $\frac{1}{1+\kappa\sigma} + \frac{\kappa}{D}$).

Ref: Benigno (2015), §8, Table 1, p. 515.

c) (2 points, T or F) At the zero lower bound with $r_n < 0$, a credible commitment to a long-run price level higher by $-r_n$ restores $y = y_n$ and $p = p^e$.

Answer: T

Step — gap formula. With $i = 0$, $y - y_n = \frac{\sigma}{1+\sigma\kappa}[r_n + (\bar{p} - p^e)] = 0$ when $\bar{p} - p^e = -r_n$; then $p = p^e$.

Ref: Benigno (2015), §9, p. 517.

d) (2 points, T or F) In the borrower–saver economy with constant inflation between the short and the long run, more price flexibility (higher κ) reduces the output cost of a deleveraging shock.

Answer: F

Step — paradox of flexibility. The AD slopes up (Fisher debt deflation); the output response $\frac{(1-\chi)\hat{d}}{\chi-(1-\chi)d_0\kappa}$ grows in size with κ (Question 4(d)).

Ref: Benigno (2015), §10, pp. 519–521.

Question 4. (15 points) Discursive. Optimal policy, the zero bound and deleveraging.

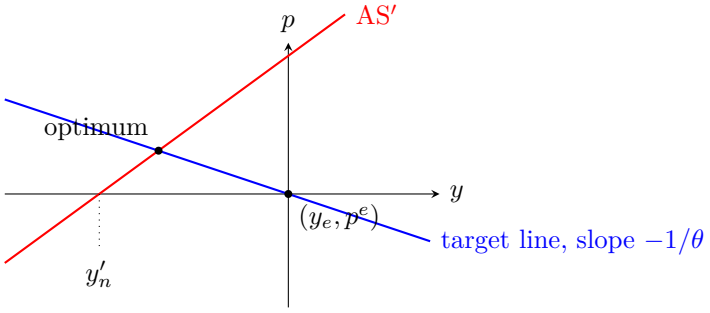
a) (3 points) The bank minimizes $L = \frac{1}{2}(y - y_e)^2 + \frac{1}{2}\frac{\theta}{\kappa}(p - p^e)^2$ (eq. 33) subject to the AS. Derive the targeting rule $(y - y_e) + \theta(p - p^e) = 0$, describe it in the (y, p) plane and prove that shocks that leave μ unchanged create no trade-off.

Step — substitution. Using $p - p^e = \kappa(y - y_n)$: $L = \frac{1}{2}(y - y_e)^2 + \frac{1}{2}\theta\kappa(y - y_n)^2$, strictly convex in y.

Step — FOC. $(y - y_e) + \theta\kappa(y - y_n) = 0$; since $\kappa(y - y_n) = p - p^e$: $(y - y_e) + \theta(p - p^e) = 0$ (eq. 34). Solution: $y^* = \frac{y_e + \theta\kappa y_n}{1+\theta\kappa}$.

Step — geometry. A line through (y_e, p^e) with slope $-1/\theta$; the optimum is its intersection with the AS.

Step — no trade-off. If μ is unchanged, $y_n = y_e$, and $y = y_e$, $p = p^e$ satisfies both the AS and the rule with $L = 0$: "divine coincidence". It is implemented by $i = r_n + (\bar{p} - p^e)$.



Schematic (a markup shock moves the AS through (y_n', p^e); θ drawn small for visibility).

Ref: Benigno (2015), §11, eqs. (33)–(34), pp. 521–522.

Grading rubric.

- Substitution and FOC: 1.5 pts.
- Geometry: 0.5 pt.
- Divine coincidence with the argument: 1 pt.

b) (4 points) For the shock of Question 2(d), compute the optimal y, $p - p^e$ and i and compare with strict price stability. Then show that no Taylor rule $i = \bar{\imath} + \psi_y(y - y_e) + \psi_\pi(p - p^e)$ with a constant intercept implements the optimum under productivity shocks: use $a = 0.004$ and $a = -0.004$.

Step — optimum. $\theta\kappa = 5.867$: $y^* = \frac{-0.0109 + 5.867(-0.0336)}{6.867} = -0.0303$; $p - p^e = \kappa(y^* - y_n) = 0.0024$; check: $(y^* - y_e) + \theta(p - p^e) = -0.0194 + 0.0194 = 0$.

Step — interest rate. From the gap formula, $i = r_n - \frac{1+\sigma\kappa}{\sigma}(y^* - y_n) = 0.0773 - 2.733 \cdot 0.0033 = 6.82\%$, between the passive 1% and strict price stability's 7.73%: the bank accepts a 0.24% price rise to cushion output.

Step — Taylor rule. Under a pure productivity shock, $y_n = y_e$ and the optimum has zero gaps, so any Taylor rule prescribes $i = \bar{\imath}$. But the optimum needs $i = r_n$: $a = 0.004$ gives $y_n = 0.0022$ and $r_n = 0.01 - 2(0.0022) = 0.56\%$; $a = -0.004$ gives $r_n = 1.44\%$. One intercept cannot equal both: the rule must track the natural rate, $i = r_n + \psi_y(y - y_e) + \psi_\pi(p - p^e)$.

Ref: Benigno (2015), §11, pp. 521–522.

Grading rubric.

- Optimal y and p, with the check: 1.5 pts.
- Optimal i and comparison: 1 pt.
- Taylor-rule argument with the two shocks: 1.5 pts.

c) (4 points) Pessimism: long-run productivity is expected to fall, $\bar{a} = -0.05$, with $a = 0$. Compute $\bar{y}_n$ and r_n. Show that the zero bound binds and compute output and prices at $i = 0$ with $\bar{p} = p^e$. What $\bar{p}$ restores the optimum? Give the limits of the outcome as $\kappa \to \infty$ and $\kappa \to 0$, and discuss credibility.

Step — natural rate. $\bar{y}_n = \frac{1.2}{2.2}(-0.05) = -0.0273$; $y_n = y_e = 0$; $r_n = 0.01 + 2(-0.0273) = -4.45\% < 0$, so $i = r_n$ is unavailable.

Step — at $i = 0$. Gap $= 0.366 \cdot (-0.0445) = -0.0163$; $p - p^e = 0.733 \cdot (-0.0163) = -0.0120$: recession and falling prices.

Step — remedy. $\bar{p} - p^e = -r_n = 4.45\%$ closes the gap and keeps $p = p^e$ (item 3(c)).

Step — limits. $\kappa \to \infty$: gap $\to 0$ and $p - p^e \to r_n = -4.45\%$, prices absorb the shock. $\kappa \to 0$: $p = p^e$ and gap $= \sigma r_n = -0.0223$.

Step — credibility. Once the recession is over, the bank prefers $\bar{p} = p^e$; the promise is time-inconsistent and works only if believed (reputation, a price-level target).

Ref: Benigno (2015), §6.3 and §9, pp. 513 and 517.

Grading rubric.

- $\bar{y}_n$, r_n and the binding bound: 1 pt.
- Outcome at $i = 0$: 1 pt.
- Required $\bar{p}$: 1 pt.
- Limits and credibility: 1 pt.

d) (4 points) With borrowers and savers (share χ of savers, initial debt d_0), Benigno's AD is

$$y = \bar{y}_n - \varphi[i - (\bar{p}_n - p) - \rho] + \tfrac{1}{\chi}(g - \bar{g}) + \tfrac{1-\chi}{\chi}[\hat{d} + d_0(p - p^e)], \qquad \varphi \equiv \frac{\sigma\chi + (1-\chi)d_0\beta}{\chi},$$

where $\hat{d}$ is the change in the debt limit. (i) Derive dy/dp when the long-run price $\bar{p}_n$ is anchored and when inflation $\bar{p}_n - p$ is constant, and name the two channels. (ii) Derive the spending multipliers in both regimes. (iii) With $\chi = 2/3$, $d_0 = 1.2$, $\beta = 0.98$, compute them for $\alpha = 0.75$ and $\alpha = 0.66$. (iv) Compute the output response to $\hat{d} = -0.1$ and explain the paradox of flexibility.

Step — slopes. Anchored: $dy/dp = -\varphi + \frac{(1-\chi)d_0}{\chi} = -\varpi$, $\varpi \equiv \sigma - \frac{d_0(1-\beta)(1-\chi)}{\chi} = 0.488$: the real-rate channel (negative) outweighs the Fisher channel (a higher p erodes nominal debt). Constant inflation: the real-rate term no longer depends on p, so $dy/dp = +\frac{(1-\chi)d_0}{\chi} > 0$: the AD slopes up.

Step — multipliers. Temporary g moves y_n by $\frac{g}{1+\sigma\eta}$ and the AD by g/χ. Anchored: $y(1+\varpi\kappa) = \frac{g}{\chi} + \frac{\varpi\kappa g}{1+\sigma\eta} \Rightarrow m = \frac{1/\chi + \varpi\kappa/(1+\sigma\eta)}{1+\varpi\kappa}$. Constant inflation: $y\big[1 - \frac{(1-\chi)d_0\kappa}{\chi}\big] = \frac{g}{\chi} - \frac{(1-\chi)d_0\kappa}{\chi}\frac{g}{1+\sigma\eta} \Rightarrow m = \frac{1-(1-\chi)d_0\kappa/(1+\sigma\eta)}{\chi-(1-\chi)d_0\kappa}$, valid when $\chi > (1-\chi)d_0\kappa$ (stability).

Step — numbers. $\alpha = 0.75$ ($\kappa = 0.733$): $m = 1.344$ (anchored), 1.964 (constant inflation). $\alpha = 0.66$ ($\kappa = 1.133$): 1.290 and 2.756, Benigno's figures. Both exceed one: Ricardian equivalence fails for constrained borrowers.

Step — deleveraging. $y = \frac{(1-\chi)\hat{d}}{\chi-(1-\chi)d_0\kappa}$ with constant inflation: -8.9% ($\alpha = 0.75$), -15.6% ($\alpha = 0.66$). Anchored: $y = \frac{(1-\chi)\hat{d}/\chi}{1+\varpi\kappa}$: -3.7% and -3.2%. More flexibility deepens the slump only when falling prices also lower expected long-run prices, raising the real value of debt.

Economic intuition. Whether flexible prices stabilize depends on what they do to expectations: with anchored long-run prices a lower p cuts the real rate; with constant inflation it inflates debt burdens.

Ref: Benigno (2015), §10, eq. (32), pp. 519–521, and footnote 20.

Grading rubric.

- Both slopes with the two channels: 1 pt.
- Both multipliers derived: 1.5 pts. Quoted from the article: max. 0.5 pt.
- Numbers: 0.5 pt.
- Deleveraging responses and the paradox, with the condition on expectations: 1 pt.